



HARSH GUPTA | 22ME31038

B.Tech.(Hons.) in MECHANICAL ENGINEERING and M.Tech. in
MANUFACTURING SCIENCE AND ENGINEERING

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EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2027	M.TECH Dual Degree 5Y	IIT Kharagpur	8.35 / 10
2021	AISSEE	DAV Bistupur	93.4%
2019	ICSE	Motilal Nehru Public School	96.8%

EXPERIENCE

Software Engineering Intern — Goldman Sachs May 2026 - Jul 2026

- Engineered **Onboarding Copilot**, a multi-agent chat platform with **8 specialist agents** querying distributed repositories
- Architected **multi-agent orchestration** using **Tree-sitter** code graphs and strict quality gates for hallucination-free routing
- Slashed investigation time from **60 to <5 minutes** and boosted confidence to **90%** via cited Root Cause Analyses

COMPETITIONS

Inter-IIT Tech Meet 13.0 — Contingent Gold Nov 2024 - Dec 2024

- Won **Gold** at **Inter-IIT Tech Meet 13.0** with BharatForge team and built a **full-stack** webapp to control ROS robots
- Developed a scalable **Flask** backend and **Next.js** frontend with noVNC and an **AI Agent** for real-time system control

Opensoft General Championship 2025 — IIT Kharagpur Mar 2025 - Apr 2025

- Led a **development team** to build a scalable **full-stack** web application with **FastAPI** and **Next.js** with live features
- Built a modular chatbot using **gRPC** and **Celery** for async processing, with focus on responsiveness and reliability

TapBonds AI Agent Competition — IIT Kharagpur Mar 2025 - Mar 2025

- Built a responsive **AI chatbot** using Next.js, and implemented intelligent AI Agents using LangChain and LangGraph
- Designed an **orchestrator system** to route queries across **four agents** for bond discovery, returns, and forecasting

PROJECTS

AI-Powered Mental Health Monitoring Platform Mar 2025 - Apr 2025

- Built an efficient mental health monitoring platform with **AI chatbot** for daily employee engagement and HR alerts
- Designed a **microservices** architecture with **gRPC**, **Celery**, **Redis**, and **MongoDB** for chat, scheduling, and updates
- Developed backend services with **FastAPI** and a **TypeScript**-based **Next.js** frontend using Socket.io and JWT auth
- Containerized** all backend and frontend services with **Docker Compose** for smooth orchestration and deployment

Automated React.js Deployment Platform Feb 2025 - Mar 2025

- Developed an **ECS Fargate** pipeline to clone GitHub React.js repositories, build bundles, and upload to **AWS S3**
- Implemented a **reverse proxy** routing subdomains to the latest **React deployment** stored in versioned S3 buckets
- Built a real-time log streaming system using **Kafka**, ClickHouse, and **Socket.io** to monitor deployments efficiently
- Enabled one-click deployments via **Express** server, provisioning build containers and triggering deployment flows

DuoBrain: Content Management Platform with Semantic Search Oct 2024 - Oct 2024

- Developed a scalable content management platform to store videos, links, and notes with efficient **semantic search**
- Built the frontend using **Next.js**, the backend using **Express** with **TypeScript**, and **MongoDB** as the main database
- Implemented **vector embeddings** for fast and efficient database indexing and enabled content sharing among users
- Dockerized** the entire project using **Docker Compose** and deployed it on **AWS EC2** using **Nginx** as a reverse proxy

AWARDS AND ACHIEVEMENTS

- Achieved the designation of **Specialist** with a maximum rating of **1462** on CodeForces (ID:harshxgupta931)
- Secured a Global rank of **327** in CodeForces Div2 (**24k+ participants**) (ID:harshxgupta931)
- Secured Global rank of **3385** in LeetCode Biweekly Contest 157 (**25k+ participants**) (ID:Harsh_Gupta_2916)
- Secured a position in the **top 1.2%** among 1 million participants in JEE Mains 2022
- Secured a position in the **top 6.8%** among 0.15 million participants in JEE Advanced 2022

COURSEWORK INFORMATION

- Computer Science:** Programming and Data Structures (Theory and Lab)
- Mathematics:** Linear Algebra, Numerical and Complex Analysis — Transform Calculus — Advanced Calculus
- MOOCs:** Algorithms and Data Structures (AlgoZenith)

SKILLS AND EXPERTISE

- Programming languages:** C++ — Python — TypeScript — Golang
- Frameworks and Libraries:** Next.js — Node.js — Express.js — Socket.io — Prisma — Mongoose — FastAPI — Flask
- Software and Tools:** Git — GitHub — Postman — MongoDB Compass — Linux (Bash, CLI)
- Devops and Cloud:** Docker — Docker Compose — Nginx — Redis — Kafka — AWS (EC2, S3, SQS, ECR, ECS)